

Krishna Vardhan Mettu

Data Scientist | Machine Learning & Predictive Modelling | Python & SQL Specialist

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Professional Summary

Data Scientist with over 4 years of experience in machine learning, predictive modelling, and ETL pipelines. Delivered scalable, cloud-based solutions such as an LLM resume matching system that cut manual screening by 70% and enhanced retrieval accuracy. Adept at using Python, AWS, and Terraform to improve data quality and drive actionable insights in Agile, collaborative environments.

Core Skills

- **Programming Language & Libraries:** Python, Pandas, NumPy, Scikit-Learn, Matplotlib, Seaborn, Plotly, R, TensorFlow, XGBoost, JavaScript, MLflow, Airflow
- **Data Science:** NLP, LLM, RAG, Feature Engineering, EDA, Fine-Tuning, ETL, A/B Testing, Hypothesis Testing, Statistical Modelling, Tableau, PowerBI, Predictive Modelling
- **Big Data & Databases:** Spark, Hadoop, Snowflake, Redshift, NoSQL, MongoDB, SQL, MySQL
- **Cloud & DevOps:** AWS, GCP, Azure, Docker, Containerization, Kubernetes, GitHub, Git, JIRA, Linux, Windows, Fast API, Rest API, CUDA, CI/CD Pipelines, MLOps Best Practices
- **Management Competencies:** Data Storytelling, Curiosity, Transparency

Professional Experience

Ocean Blue Solutions

Jun 2025 - Present

Data Scientist | Python, AWS, Gen AI, RAG, Terraform

Houston, TX, USA

- Developed and implemented an LLM-based resume-job description matching system achieving **92% accuracy**, leveraging advanced data analysis techniques to extract meaningful insights from text.
- Preprocesses and stored **5,000+ resumes** in an **S3 bucket** and indexed them in a **vector database**, streamlining data integration and enhancing query response times by **65%**.
- Converted job descriptions into vector embeddings for semantic search and NLP-based retrieval, **effectively reducing manual resume screening efforts by 70%** and incorporating machine learning methods in data cleansing.
- Engineered a serverless application on AWS Lambda deployed via Terraform, **reducing cloud infrastructure costs by 40%** while ensuring automated, reproducible deployments in an Agile environment.
- Automated candidate shortlisting workflows using IaC, enabling recruiters to identify top resumes in under 10 seconds and **improving hiring efficiency by 3x** while clearly communicating technical solutions to stakeholders.

G-Think Inventors

Apr 2020 - Jul 2023

Data Scientist | Predictive Modelling, RNN, Scikit-learn, ETL, SQL, AWS, Python, Energy

Hyderabad, India

- Designed interactive dashboards and automated reports using predictive models to visualize appliance usage and energy savings, enabling customers to **reduce electricity bills by 25%** and demonstrating strong analytical abilities.
- Built ETL workflows with Apache Spark and Python (Pandas, NumPy) to preprocess high-volume time-series data, **improving data quality by 30%** and integrating statistical programming techniques into CI/CD pipelines.
- Optimized SQL queries and refined database schemas to streamline data retrieval and reporting, **reducing query execution time by 40%** and facilitating timely decision-making for both technical and non-technical audiences.
- Implemented scalable MLOps practices by deploying analytics infrastructure on AWS (Lambda, SageMaker, S3), **boosting data processing scalability by 34% and ensuring 99.9% uptime** under rigorous testing conditions.

- Collaborated with cross-functional stakeholders, including data scientists, engineers, and consultants, to document component features, ensuring aligned execution across teams and adherence to Agile methodologies.

G-Think Inventors

Jun 2019 - Apr 2020

Internship | Python, API, AWS, Data Extraction, Energy

Hyderabad, India

- Built and deployed an ETL pipeline to extract live sensor data from appliances via REST APIs in JSON format, **enhancing data consistency by 40%** and demonstrating practical predictive modelling support.
- Implemented preprocessing steps including noise removal, duplicate handling, and normalization that elevated **data quality by 35%**, thereby improving the accuracy of subsequent analytics and visualizations.
- Designed and automated data transformation workflows converting raw JSON into CSV format, **reducing data preparation time by 50%** and enabling efficient statistical analysis.
- Engineered secure and scalable data storage solutions on AWS S3, enabling rapid data retrieval and **reducing query latency by 45%** for downstream analytical tasks.
- Authored detailed technical documentation and process guidelines, **reducing onboarding time by 60%** and fostering clearer communication with both technical and non-technical stakeholders.

Education

University of Central Missouri, USA

Aug 2023 - May 2025

Masters, Big Data Analytics and Information Technology

Warrensburg, MO, USA

Sreyas Institute of Engineering, India

Jun 2016 - Apr 2020

Bachelors, Computer Science and Engineering

Hyderabad, India

Projects

House Affordability and Price Prediction | Python, MySQL, Flask, Power BI, Scikit-learn

- I developed a data ingestion pipeline using Fast API and MySQL, processing and cleansing the data with Python (Pandas, NumPy), which **improved data quality by 90%**.
- I built an analytics engine with Scikit-learn predictive models for housing price prediction, **achieving 92% accuracy**.
- Created an interactive dashboard with Power BI and Flask, **reducing report generation time by 85%**, and engineered a full-stack platform that **increased user engagement by 50%**.

Financial Research Tool | Python, LLM, RAG, Lang chain, OpenAI, FAISS

- Designed and implemented a data-driven financial research tool using LLMs and RAG, which automated data extraction and analysis, **cutting manual research effort by 70%** and providing rapid insights for financial decision-making.
- Optimized data chunking and embedding in a vector database, boosting retrieval speed by 3x and **reducing OpenAI API costs by 40%**.

Certification

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- **NVIDIA-Certified Associate:**Generative AI LLMS
 - **Tvastaa Data Solution:**Python Programming in Data Analytics

Publication

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- Shaik Abdul Nabi & Mettu Krishna Vardhan (2021). **NavRobotVAC: a navigational robotic vacuum cleaner using Raspberry Pi and Python**. In Smart innovation, systems and technologies (pp. 159–167). https://link.springer.com/chapter/10.1007/978-981-16-1502-3_17